

*MA.7.NSO.2.3***Benchmark**

MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.

Benchmark Clarifications:

Clarification 1: Instruction includes using one or more operations to solve problems.

Connecting Benchmarks/Horizontal Alignment

- MA.7.AR.2.1, MA.7.AR.2.2
- MA.7.AR.3.1, MA.7.AR.3.2, MA.7.AR.3.3
- MA.7.AR.4.1, MA.7.AR.4.2, MA.7.AR.4.3, MA.7.AR.4.4, MA.7.AR.4.5
- MA.7.GR.1.1, MA.7.GR.1.2, MA.7.GR.1.3
- MA.7.GR.2.1, MA.7.GR.2.2, MA.7.GR.2.3
- MA.7.DP.1.1, MA.7.DP.1.2, MA.7.DP.1.3

Terms from the K-12 Glossary

- Rational Number

Vertical Alignment**Previous Benchmarks**

- MA.6.NSO.2.3

Next Benchmarks

- MA.8.NSO.1.6, MA.8.NSO.1.7

Purpose and Instructional Strategies

In grade 6, students solve real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers. In grade 7, students move to all rational numbers and in grade 8 extend to rational numbers, including exponents and radicals.

- This benchmark applies the procedural fluency skills of the previous benchmark to real-world problems (*MTR.3.1*).
- Students should develop fluency with and without the use of a calculator when performing operations with rational numbers.
- Instruction includes the use of technology to help emphasize the proper use of grouping symbols for order of operations.
- Students should have experience using technology with decimals and fractions as they occur in the real world. This experience will help to prepare students working with irrational numbers in grade 8.
- Open-ended tasks with real-world contexts (*MTR.7.1*) will allow students to practice multiple pathways for solutions as well as to make comparisons with their peers (*MTR.4.1*) to refine their problem-solving methods.
- Instruction includes support in vocabulary development as related to the context of real-world problems.

Common Misconceptions or Errors

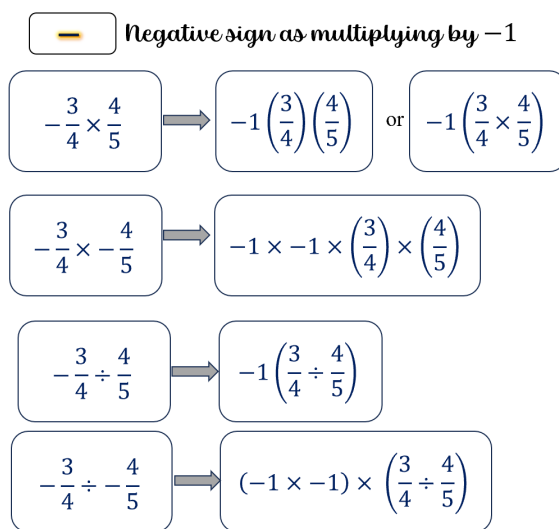
- Students may incorrectly perform operations with the numbers in the problem based on what has recently been taught, rather than what is most appropriate for a solution. To overcome this misconception, have students estimate or predict solutions prior to solving

them and then compare those predictions to their actual solution to see if it is reasonable (*MTR.6.1*).

- Students may incorrectly oversimplify a problem by circling the numbers, underlining the question, boxing in key words, and eliminating contextual information that is needed for the solution. This process can cause students to not be able to comprehend the context or the situation (*MTR.2.1, MTR.4.1, MTR.5.1, MTR.7.1*).

Strategies to Support Tiered Instruction

- Instruction includes the use of visual representations and manipulatives to represent the given situation and use the chosen representation to help find the solution.
- The teacher provides opportunities for students to comprehend the context or situation by engaging in questions like the ones below.
 - What do you know from the problem?
 - What is the problem asking you to find?
 - Can you create a visual model to help you understand or see patterns in your problem?
- Teacher co-creates a graphic organizer with students to review operations with positive fractions and operations with integers to assist when applying operations with rational numbers.



- Instruction includes the use of a three-read strategy. Students read the problem three different times, each with a different purpose. Laminating these questions on a printed card for students to utilize as a resource in and out of the classroom would be helpful.
 - First, read the problem with the purpose of answering the question: What is the problem, context, or story about?
 - Second, read the problem with the purpose of answering the question: What are we trying to find out?
 - Third, read the problem with the purpose of answering the question: What information is important in the problem?
- Instruction includes having students estimate or predict solutions prior to solving them and then compare those predictions to their actual solution to see if it is reasonable.

Instructional Tasks

Instructional Task 1 (MTR.7.1)

All of the 7th grade homeroom classes collected recycling, with the top three classes splitting the grand prize of \$800 toward building their own gardens. Mr. Brogle's class turned in 237 pounds of recycling, Mrs. Abiola's class turned in 192 pounds and Mr. Wheeler's class turned in 179 pounds. How should these top three divide the money so that each class gets the same fraction of the prize money as the fraction of recycling they collected?

Instructional Task 2 (MTR.7.1)

Kari and Natalia went to the Fun Warehouse with \$20 each to spend. There is a \$3 entry fee each and the menu of activities is shown below. What are some possible combinations of activities Kari and Natalia can enjoy before they each run out of money?

Go-Karts	\$9.75
Laser Tag	\$7.50
Inflatables	\$5.00
Mini-Bowling	\$2.25
Video Games	\$0.75

Instructional Task 3 (MTR.4.1, MTR.7.1)

Anjeanette is making cupcakes for her sister's birthday. Among other ingredients, her recipe calls for 2 cups of flour, $\frac{1}{2}$ cup of butter and $\frac{3}{4}$ cup sugar in one batch. In the kitchen, she has 8 cups of flour, 2 cups of butter and 2 cups of sugar.

Part A. How many batches of cupcakes can Anjeanette make?

Part B. What should Anjeanette ask for if she wants to borrow from her neighbor to make one more batch?

Instructional Items

Instructional Item 1

Kari and Natalia went to the Fun Warehouse with \$20 each to spend. They paid the \$3 entry fee each and then decided they would both play laser tag and mini-bowling. If they finished the day with playing 4 video games each, how much money will be left?

Go-Karts	\$9.75
Laser Tag	\$7.50
Inflatables	\$5.00
Mini-Bowling	\$2.25
Video Games	\$0.75

**The strategies, tasks and items included in the BIG-M are examples and should not be considered*